



Chapter 5

Image restoration

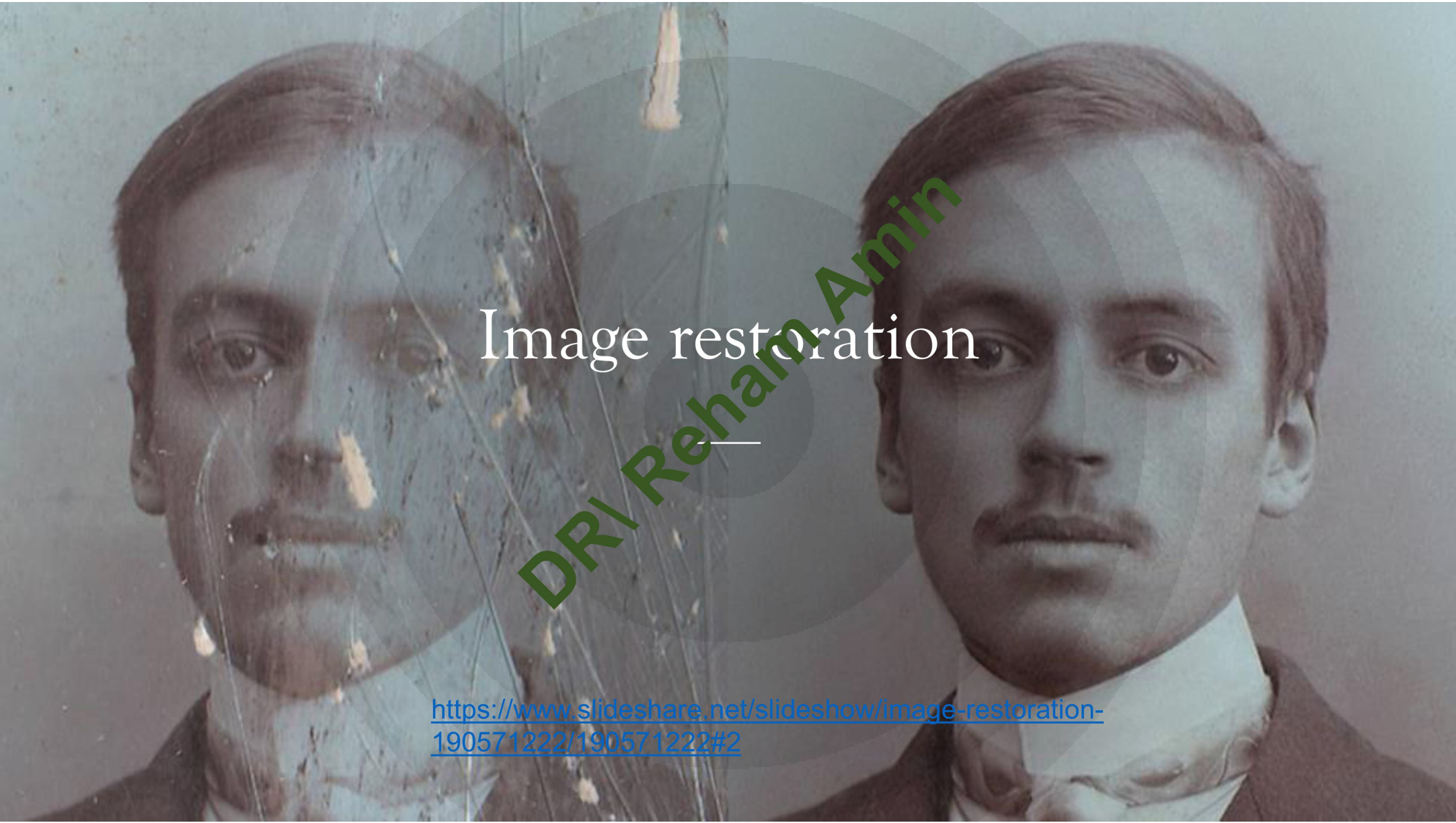


Image restoration

<https://www.slideshare.net/slideshow/image-restoration-190571222/190571222#2>

Image restoration

- The objective of restoration is to improve the quality of a digital image which has been degraded due to Various phenomena like:
 - Motion
 - Improper focusing of Camera during image acquisition.
 - Noise Modified from restoration
- Image restoration is to restore a degraded/distorted image to its original content and quality.
- Restoration involves following process:-
 - Modeling of Degradation
 - Applying the inverse process to recover the original image

Image restoration in spatial and frequency domains

In this chapter, we model image degradation as an operator $\mathcal{H}$ that, together with an additive noise term, operates on an input image $f(x, y)$ to produce a degraded image $g(x, y)$ (see Fig. 5.1). Given $g(x, y)$, some knowledge about $\mathcal{H}$, and some knowledge about the additive noise term $\eta(x, y)$, the objective of restoration is to obtain an estimate $\hat{f}(x, y)$ of the original image. We want the estimate to be as close as possible to the original image and, in general, the more we know about $\mathcal{H}$ and η , the closer $\hat{f}(x, y)$ will be to $f(x, y)$.

We will show in Section 5.5 that, if $\mathcal{H}$ is a linear, position-invariant operator, then the degraded image is given in the spatial domain by

$$g(x, y) = (h \star f)(x, y) + \eta(x, y) \quad (5-1)$$

where $h(x, y)$ is the spatial representation of the degradation function. As in Chapters 3 and 4, the symbol “ $\star$ ” indicates convolution. It follows from the convolution theorem that the equivalent of Eq. (5-1) in the frequency domain is

$$G(u, v) = H(u, v)F(u, v) + N(u, v) \quad (5-2)$$

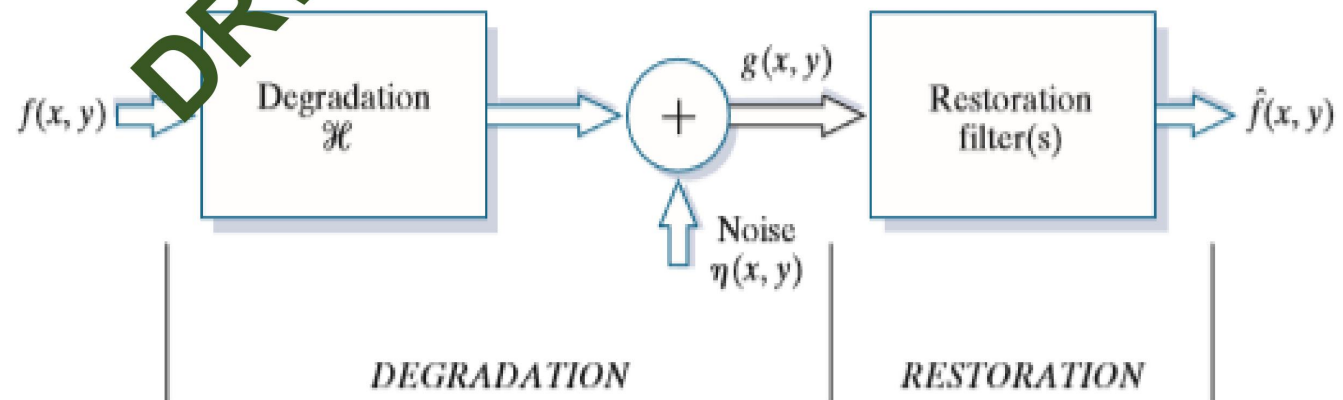
where the terms in capital letters are the Fourier transforms of the corresponding terms in Eq. (5-1). These two equations are the foundation for most of the restora-

Model of image restoration

- We will assume that a degradation function exists, which, together with additive noise, operates on the input image $f(x,y)$ to produce a degraded image $g(x,y)$.
- The objective of restoration is to obtain an estimate for the original image from its degraded version $g(x,y)$ while having some knowledge about the degradation function H and the noise

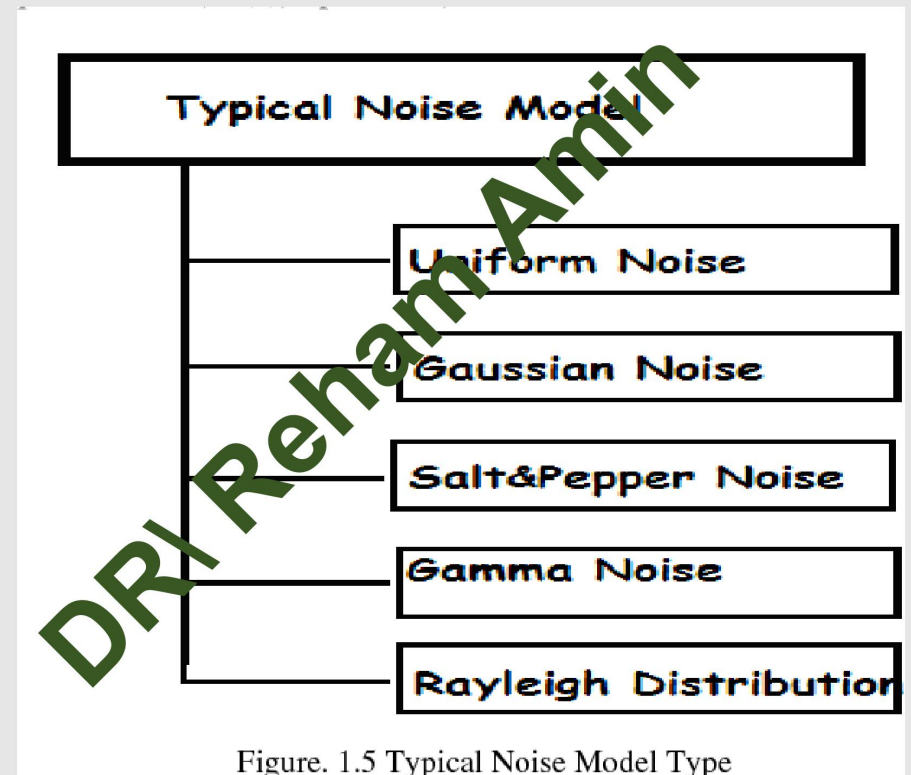
FIGURE 5.1

A model of the image degradation/restoration process.



Noise source

- The Principal source of noise in digital images arise:
 - During image Acquisition and transmission.
 - In Acquiring images with Camera and Light levels are major factor affecting the amount of noise in resulting image.
 - Images are corrupted during transmission due to interferences in the channel used for transmission.
- for example: image is transmitted using a wireless network might be corrupted due to atmospheric disturbances.
- Some important noise probability density functions
 - Gaussian noise
 - Rayleigh noise
 - Erlang gamma noise
 - Exponential noise
 - Impulse



Uniform noise

The PDF of *uniform* noise is

$$p(z) = \begin{cases} \frac{1}{b-a} & a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

(5-13)

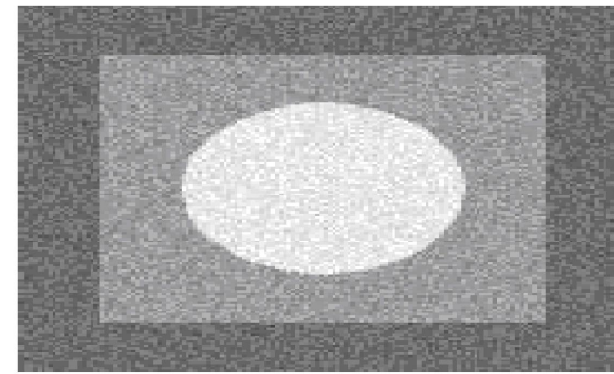
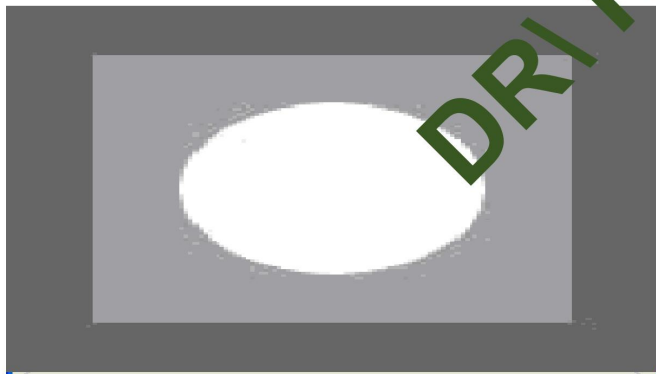
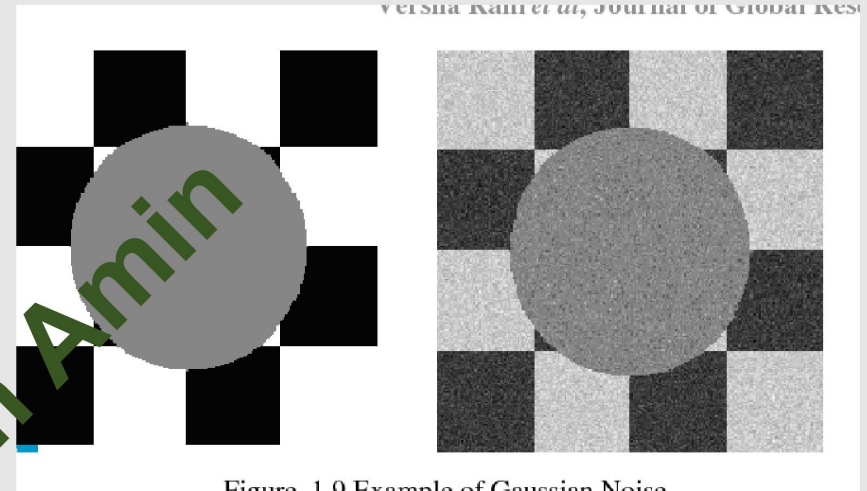


Figure. 1.7 Example of Uniform Noise

Gaussian Noise or Amplifier Noise:

Gaussian Noise

- The Gaussian noise arises in an image due to factors such as electronic circuit noise and sensor noise due to poor illumination.
- The images acquired by image scanners exhibit this phenomenon.



The PDF of a *Gaussian* random variable, z , is defined by the following familiar expression:

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z - \bar{z})^2}{2\sigma^2}} \quad -\infty < z < \infty \quad (5-3)$$

where z represents intensity, $\bar{z}$ is the mean (average) value of z , and σ is its standard deviation. Figure 5.2(a) shows a plot of this function. We know that for a Gaussian

Salt and pepper noise



If k represents the number of bits used to represent the intensity values in a digital image, then the range of possible intensity values for that image is $[0, 2^k - 1]$ (e.g., $[0, 255]$ for an 8-bit image). The PDF of *salt-and-pepper* noise is given by

$$p(z) = \begin{cases} P_s & \text{for } z = 2^k - 1 \\ P_p & \text{for } z = 0 \\ 1 - (P_s + P_p) & \text{for } z = V \end{cases} \quad (5-16)$$

where V is any integer value in the range $0 < V < 2^k - 1$.

Erlang (gamma) noise

Erlang (Gamma) noise:

$$p(z) = \begin{cases} \frac{\alpha^b z^{b-1}}{(b-1)!} e^{-\alpha z} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

$$\mu = b / \alpha;$$

$$\sigma^2 = b / \alpha^2$$

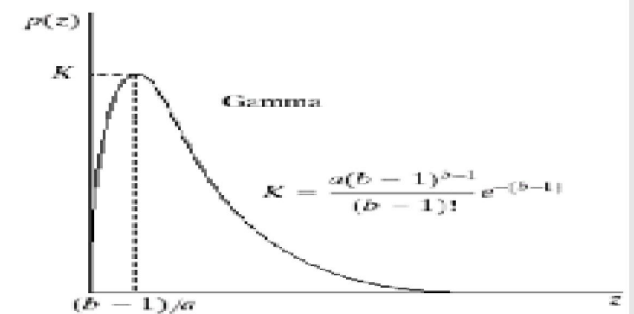


Figure. 1.13 PDF, Mean, Variance of Gamma Noise

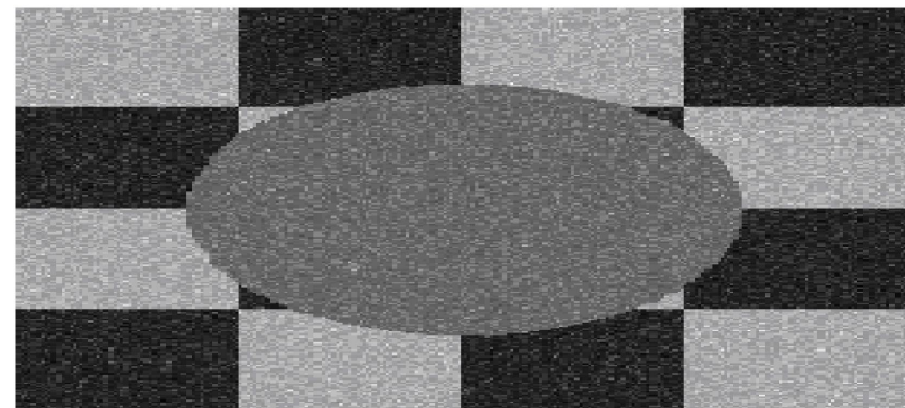
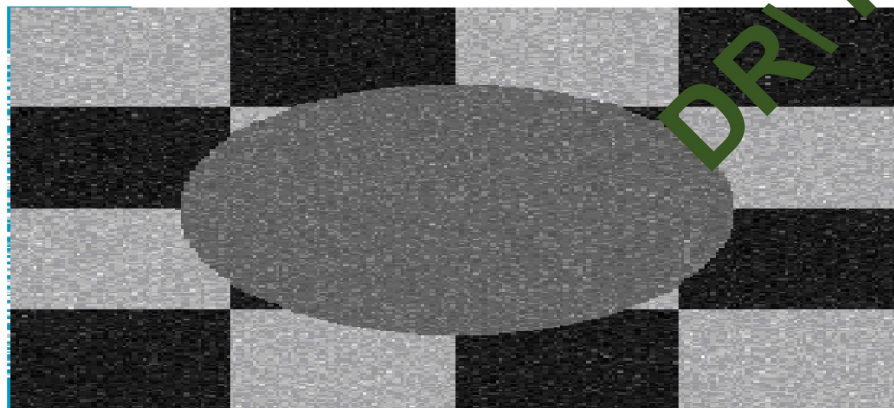


Figure. 1.14 Example of Rayleigh Noise and Gamma Noise

Exponential Noise

The PDF of *exponential* noise is given by

$$p(z) = \begin{cases} ae^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases} \quad (5-10)$$

where $a > 0$. The mean and variance of z are

$$\mu = \frac{1}{a} \quad (5-11)$$

and

$$\sigma^2 = \frac{1}{a^2} \quad (5-12)$$

Note that this PDF is a special case of the Erlang PDF with $b = 1$. Figure 5.2(d) shows a plot of the exponential density function.

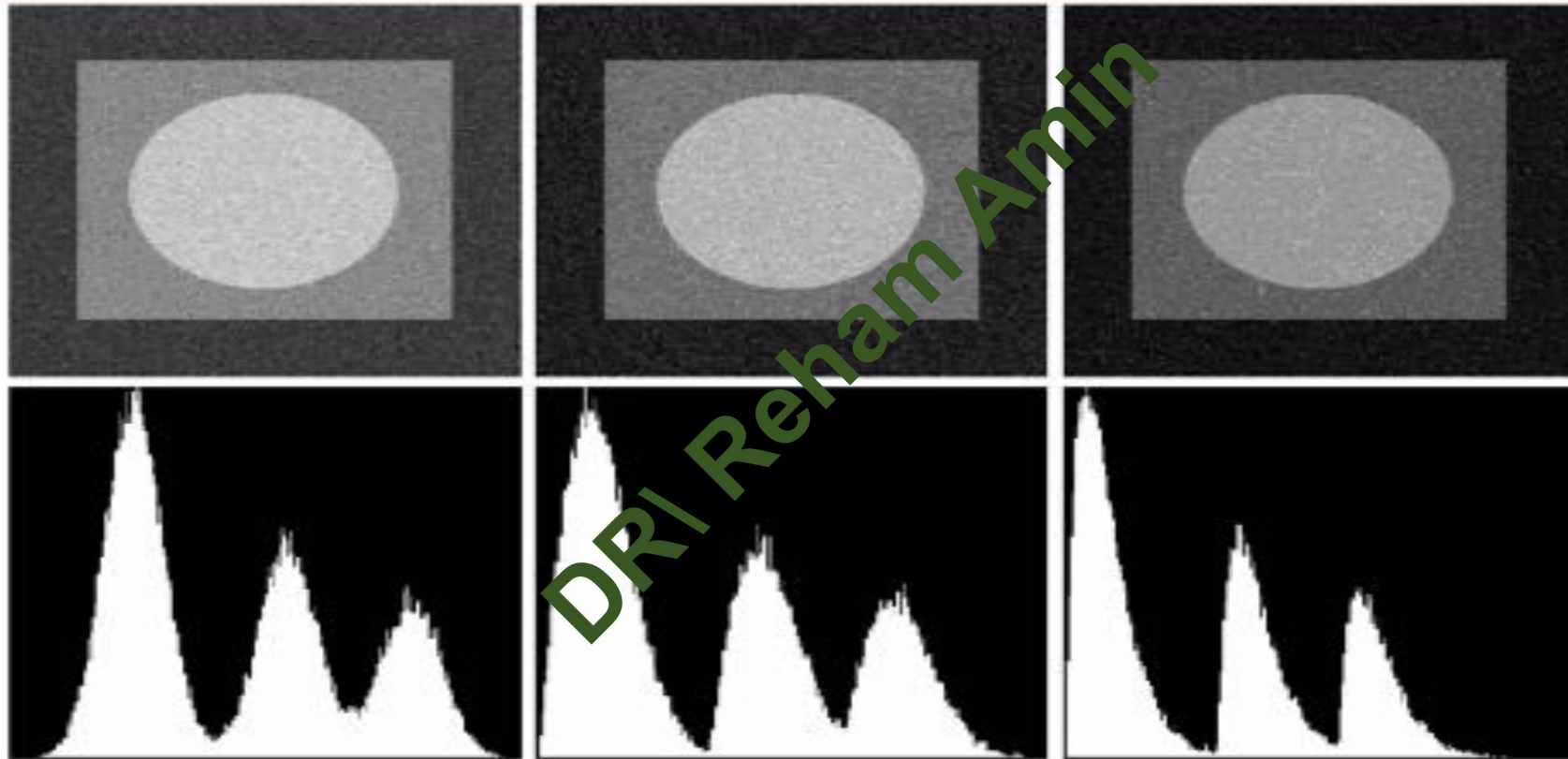
It can be found in laser imaging.

Effect of Noise on Images & Histograms

Test pattern used to illustrate the noise models → areas of 3 constant gray levels (white, gray, black)



Effect of Noise on Images & Histograms

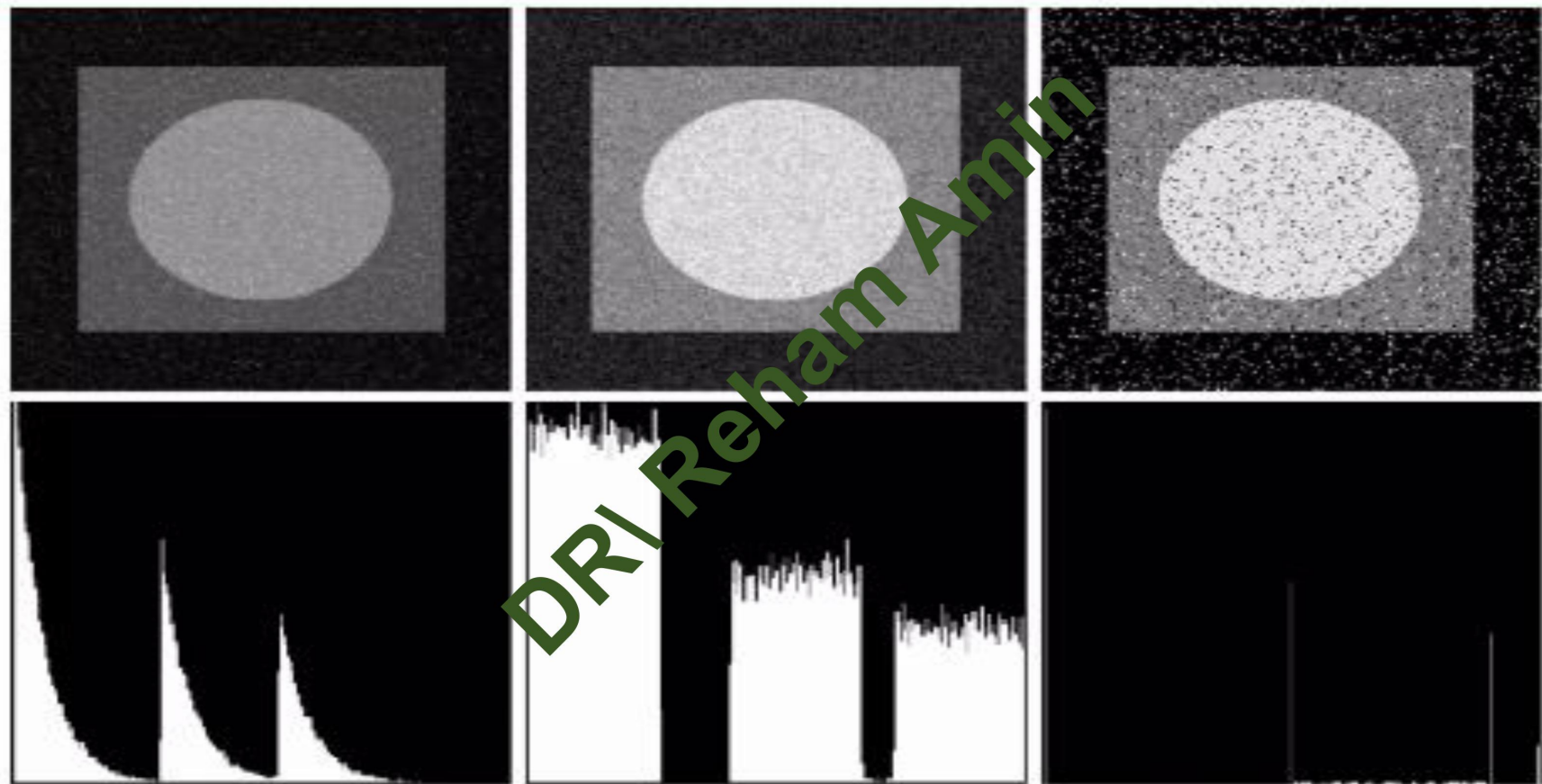


Gaussian

Rayleigh

Gamma

Effect of Noise on Images & Histograms



Exponential

Uniform

Salt & Pepper

Noise removal restoration methods in spatial domain

- [□ Mean filters >](#)
 - Arithmetic mean filter
 - Geometric mean filter
 - Harmonic mean filter
 - Contra-harmonic mean filter
- Order statistics filters
 - Median filter
 - Max and min filters
 - Mid-point filter
 - alpha-trimmed filters
- Adaptive filters
 - Adaptive local noise reduction filter
 - Adaptive median filter

⦿ Arithmetic mean filter:

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(s, j) \in S} g(s, j)$$

The arithmetic mean filter computes the average value of the corrupted image $g(x, y)$ in the area defined by S_{xy} . Let S_{xy} represent the set of coordinates in a rectangular neighborhood of size $m \times n$, centered at the point (x, y) .

Effect: The Mean filter simply smoothes the variations in an image. noise is reduced as a result of blurring

Consider the following image partition, apply the mean filter to the center value

8	4	7
2	1	9
5	3	6

$$8+4+7+2+1+9+5+3+6=45$$

$$45 / 9 = 5$$

Figure. 1.16 An Example of mean filtering of a 3x3 kernel of values is shown below

Mean filtered		
*	*	*
*	5	*
*	*	*

Figure. 1.17 In this Center value which is previously 1 in the unfiltered value is replaced by the mean of all nine values that is 5.

Advantage:

- Easy to implement
- Used to remove the impulse noise.

Geometric mean filter:

$$\hat{f}(x, y) = \left[\prod_{(s,t) \in \Omega} g(s, t) \right]^{\frac{1}{|\Omega|}}$$

Effect:

Geometric mean filter achieves smoothing comparable to the arithmetic mean filter but it preserves more details (It means loss less image detail in the process)

Harmonic mean filter:

$$\hat{f}(x, y) = \frac{mn}{\sum_{(s,t) \in S_{xy}} \frac{1}{g(s,t)}}$$

Effect:

Harmonic mean filter works well for salt noise and other types of noise (such as Gaussian) but fails for pepper noise.

Noise removal restoration methods in spatial domain

- [□ Mean filters >](#)
 - Arithmetic mean filter
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Order-statistic filters

	10	5	20				
	14	80	11				
	8	3	22				

3, 5, 8, 10, 11, 14, 20, 22, 80

median (central value 80 is replaced by 11)

Max and Min Filters

⊙ Max filters:

$$\hat{f}(x, y) = \max_{(s, t) \in S_w} \{g(s, t)\}$$

This filter is useful for finding the brightest points in an image; therefore, its effective against pepper noise.

Min filters:

$$\hat{f}(x, y) = \min_{(s, t) \in S_w} \{g(s, t)\}$$

This filter is useful for finding the darkest points in an image; therefore, its effective against salt noise. (it reduces the salt noise because it will eliminate the higher gray values in the subimage).

A photograph of a vast, hazy landscape. In the foreground, there are jagged, reddish-brown rock formations. Beyond them is a large body of water, possibly a lake or a wide river, which appears calm and reflects the light. The background is a hazy, mountainous or forested area under a soft, overcast sky. The overall color palette is muted, with earthy tones and soft blues and greys.

Sheet time

Sheet

Prepare mathematical examples (at least 3 different examples for each one) for

- Geometric mean filter
- Max filter
- Min filter
- Median filter

The image shows a chalkboard with handwritten mathematical derivations. On the left, a graph of a function $y = g(x)$ is shown with a point T and a tangent line. The x-axis is labeled $x+h$. The text "Secant Lines" and "Tangent Line" is written near the graph. On the right, the derivative $f'(x)$ is defined as the limit of the difference quotient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Below this, the derivative is calculated for $f(x) = x^2$:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} h(2x + h) \end{aligned}$$



Handwritten mathematical derivations on a chalkboard:

$y = g(x)$

Secant Lines

Tangent Line

$x+h$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$
$$f(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{h(2x + h)}{h}$$
$$= \lim_{h \rightarrow 0} (2x + h)$$

Sheet
using MATLAB, Implement the mathematical examples
(using 3 different images)
for :



Geometric mean filter



Max filter



Min filter

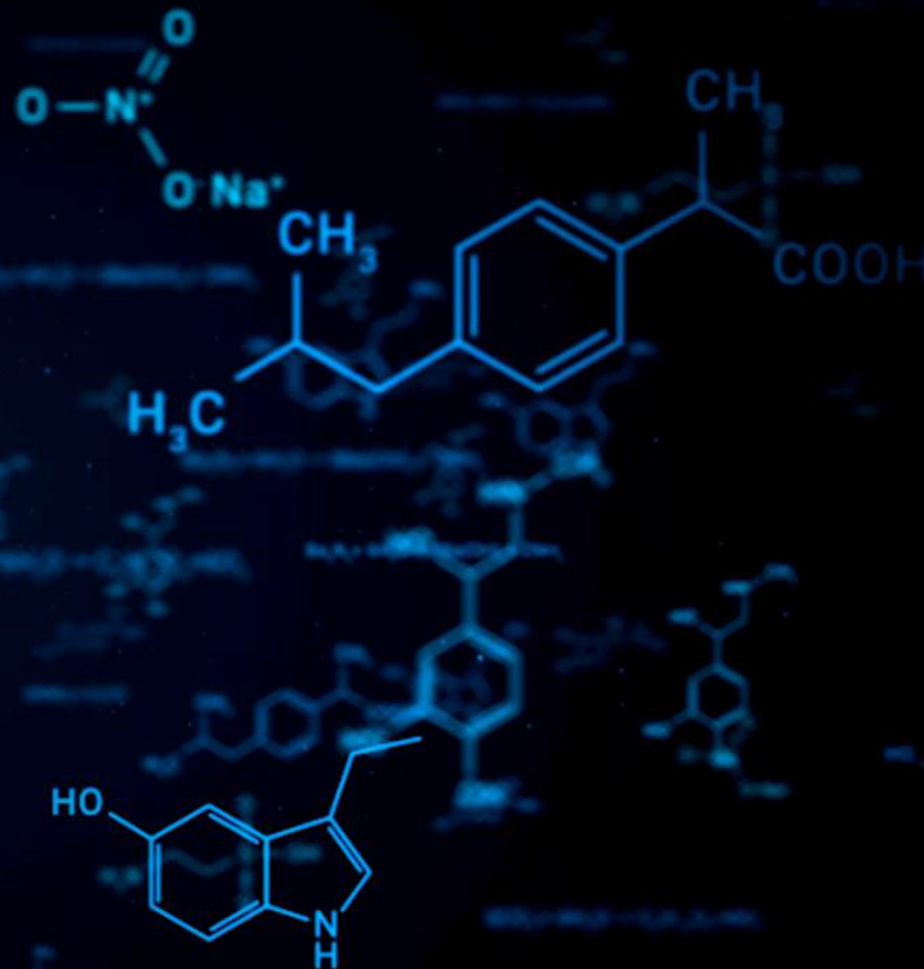


Median filter

Adaptive filters

- Adaptive filters are those filters whose behavior changes based on the statistical characteristics of the image inside the filter region defined by a rectangular window size.
 - It is better than the mean filter and order statistics filter.
 - Two types of filter
 - › Adaptive local noise reduction filter
 - › Adaptive median filter

Periodic noise by Frequency domain filtering



Periodic noise by Frequency domain filtering

- This noise typically comes from electrical or electromechanical interference during image acquisition.
- It can be reduced via frequency domain filtering.
- The image is corrupted by sinusoidal noise of various frequencies.
- The parameters of periodic noise are estimated by inspection of Fourier spectrum of the image.

Periodic noise reduction filters

- Bandreject filters
- Bandpass filters
- Notch Filters

Bandreject filters

- Bandreject filters remove or attenuate frequencies about the origin of the Fourier transform.
- Ideal Bandreject Filter:

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) < D_0 - W/2 \\ 0 & \text{if } D_0 - W/2 \leq D(u,v) \leq D_0 + W/2 \\ 1 & \text{if } D(u,v) > D_0 + W/2 \end{cases}$$

Where

$D(u,v)$: distance from the origin of the centered freq.

D_0 : Radial centre

W - width of the band

A photograph of a vast, hazy landscape. In the foreground, there are jagged, reddish-brown rock formations. Beyond them is a large body of water, possibly a lake or a wide river, which appears calm and reflects the light. The background is a hazy, mountainous or forested area under a soft, overcast sky. The overall color palette is muted, with earthy tones and soft blues and greys.

Sheet time

Sheet

Prepare mathematical examples (at least 3 different examples for each one)

for

- Band reject filter

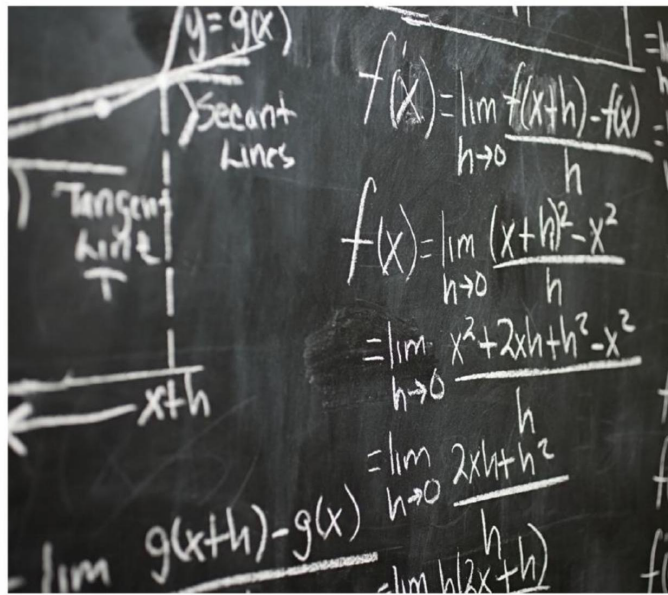
The image shows a chalkboard with handwritten mathematical derivations. On the left, a graph of a function $y = g(x)$ is shown with a point $x+h$ on the x-axis. A secant line is drawn through the point $(x, g(x))$ and $(x+h, g(x+h))$, labeled "Secant Lines". A tangent line is also shown, labeled "Tangent Line".

On the right, the derivative $f'(x)$ is defined as the limit of the difference quotient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

For $f(x) = x^2$, the derivative is calculated as follows:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} h(2x + h) \end{aligned}$$



Sheet
using MATLAB, Implement the mathematical
 example
 (using 3 different images)

for:

- Band reject filter

DRI Reham Amin

Thank you

